

Pavel Ovchinnikov

Full-Stack Web Developer

BASED IN Georgia (PE)

ENGLISH B2



MY EXAMPLES



Meshintex

Nextjs, React, Tailwind



GraphQL boilerplate

TypeGraphQL, TypeORM, Apollo, NextJS



AD Dashboard

Vite, React, Sass



PWA Mobile app

Ionic, Angular, Google Maps API



NFT Marketplace

React, GraphQL, Sass



Group chat app

React, Socket.io, MUI, GCP, e2e tests



Storybook boilerplate

Storybook, NPM, React, Sass

Full-stack web developer with 10+ years of experience building web and mobile applications. Comfortable across the entire stack — from database design and backend APIs to frontend UIs.

My recent achievements:

- IoT service (website + dashboard) in 20 hours from scratch to prod (CRUD, SEO, charts, sensor map, CI/CD, SOC2 ready) — meshintex.com
- ERP/CMMS system in 2 business days from scratch to prod (inventory, invoices, orders, payment processing, CI/CD, ghcr)

I have a deep expertise in **React** (hooks, RSC) with **Next.js** for full-stack and SSR/SSG/ISR workflows. **Vue** experience including **Nuxt**. Production **Angular** apps with **RxJS**, reactive forms. Solid state management across ecosystems (**Redux Toolkit**, **Zustand**) on the React side; **Vuex** for Vue; **NgRx** for Angular. Graphics work using **SVG** (animations, data viz, custom UI) and **Canvas** (real-time rendering, image manipulation). Real-time UX via sockets and SSE. Styling & UI: CSS-in-JS, **Tailwind**, SCSS/CSS Modules, fluid typography, custom animation with **Framer Motion** and CSS transitions. Built component libraries using **Storybook**. Use **Vite**, **Webpack**. Experienced optimizing bundle size by code splitting, tree-shaking, lazy loading, and analyzing bundle output. Testing: Unit and integration tests with **Mocha**, **Jest/Vitest**. E2E with **Cypress** and **Puppeteer**. Visual regression testing.

Develop APIs, data models, and system architecture. Production experience with high-load systems built on microservices — including an SNS/SQS + Lambda pipeline processing 30k health-check requests per minute. **PostgreSQL** as the primary relational database — schema design, query optimization, indexing strategies, migrations. ORM experience with **TypeORM** and **Sequelize** across multiple production codebases. **DynamoDB** for high-performance, low-latency workloads. Builds APIs in REST (with **OpenAPI/Swagger** schemas as the contract) and **GraphQL** (schema design, resolvers, subscriptions). Production experience with multiple auth approaches: **OpenID**, **Auth0**, **AWS Amplify** (Cognito-based auth), and **next-auth** with different providers.

AWS (S3, EC2, Lambda, Cognito, SNS, RDS, CloudFront) and **Google Cloud**. CI/CD pipelines, automated testing, infrastructure-as-code. Experienced deploying and scaling services in cloud-native environments.

Shipped production mobile apps using **React Native**, **Flutter**, and **Ionic** — native-quality experiences from a shared codebase.

Maps and geospatial integrations (**Google Maps**, **Mapbox**, **Leaflet**) in production.

Mentors and leads developers formally and informally. Fluent with AI-assisted development tools with a clear methodology for using them to ship faster without losing quality or security. Reviews teammates' code to catch bugs and keep quality high. Writes clear docs, branching policies, and linting rules. Good at detecting issues early using the terminal, curl, logs, and native database queries.

2022_{.07} - to date



At Kupsilla I worked on two client projects: **Strateos**, a cloud lab automation platform, and a **Genetic Science** research application.

Conducted **technical interviews** for junior and mid-level candidates, performed **code reviews**, and **mentored** developers to keep code quality high and support team growth.

Strateos – A **Cloud Lab Automation-as-a-Service** platform where users remotely run real **chemical reactions via robotic systems**.

Improved scalability and independent deployability of a large React codebase, measured by separate release cycles per team, by re-architecting the frontend into **microfrontends** using **React**, **TypeScript**, **SCSS**, and **Storybook**. ↑ **SCALABILITY**

Increased development speed across teams, measured by reduced duplication of UI work, by building shared UI libraries and publishing them to a private **NPM registry** used by all microfrontend modules. ↑ **DEV SPEED**

Built a complex chemical reaction constructor — a heavy custom UI component that lets users visually design lab reactions — and packaged it as a standalone NPM module to keep the main app lightweight and maintainable. ↑ **MAINTAINABILITY**

Genetic Science – A web application for searching and visualizing scientific parameters related to **human genes**.

Reduced initial page load time for data-heavy genomic pages by implementing **SSR** and

enabling **gzip compression** on API responses. ↑ **PERFORMANCE**

Kept AWS infrastructure costs within the free tier, measured by zero hosting spend, by deploying on **AWS Amplify** and managing build-minute usage within free quota limits.

↓ **COST**

Built a complex interactive view combining a data plot and a collapsible rows table in one synchronized layout using **nivo.rocks**, improving data exploration for researchers. ↑ **UX**

Enabled smooth collaboration across the team, measured by a consistent CI/CD flow for all developers, by setting up **AWS Amplify** with **Amazon Cognito** (Google OAuth), linting, testing, and a **GitHub branching policy** from scratch. ↑ **TEAM FLOW**

2021.05 - 2022.07

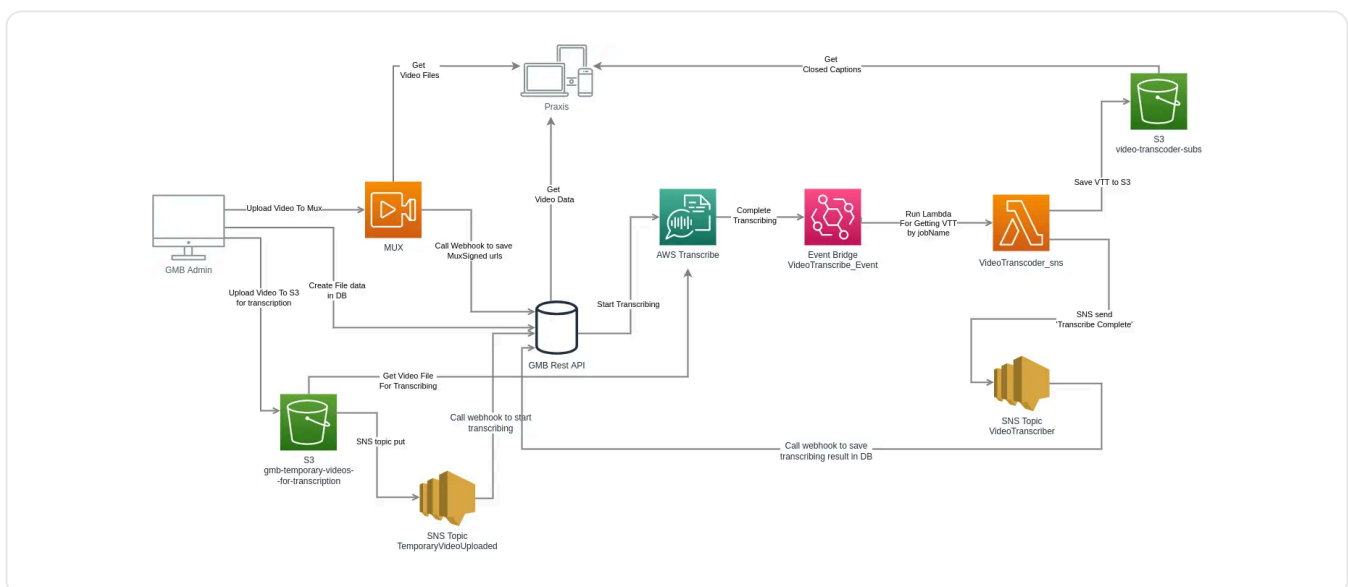


At Strata I worked on **gmb.io** and **Praxis** — a fitness video training platform for North American users. Stack: **Next.js**, **NestJS**, **PostgreSQL**, **TypeORM**, **Auth0**, **AWS Lambda**, **MUX**.

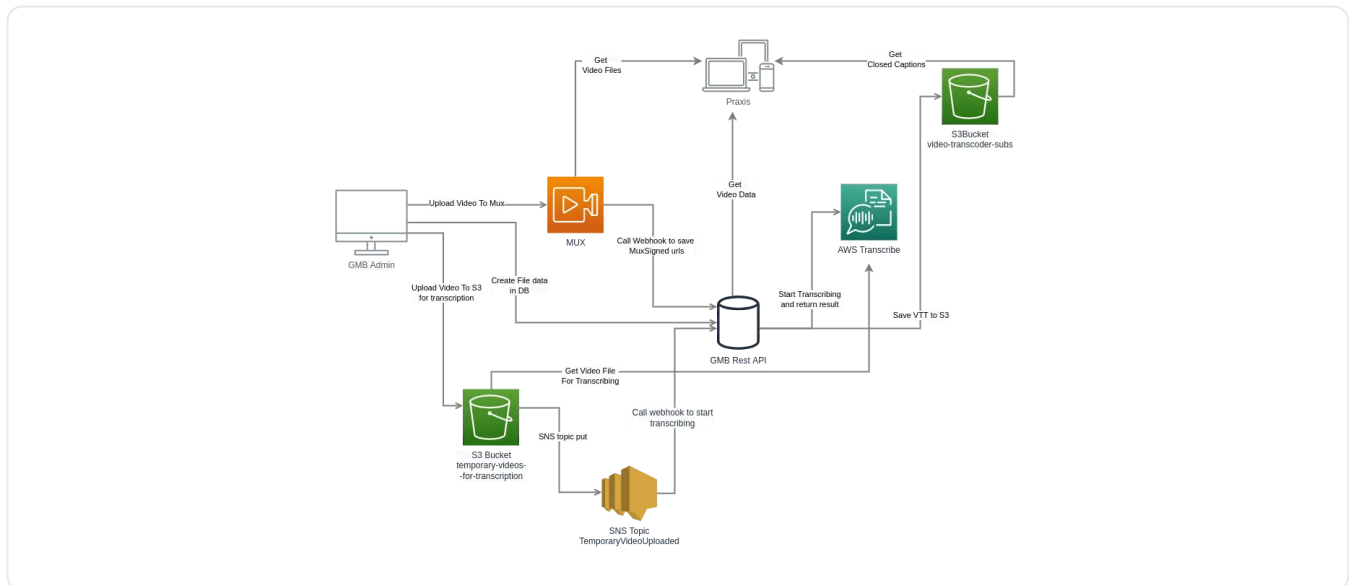
Improved video streaming quality across a wide range of devices, measured by reduced buffering and format errors, by implementing automatic platform detection that selects the optimal video codec and resolution for each client. ↑ **VIDEO QUALITY**

Reduced the maintenance cost of the video upload pipeline, measured by the number of AWS services needed to operate it, by simplifying the microservice architecture and removing redundant steps. ↓ **OPS COST**

PREVIOUS FLOW



SIMPLIFIED FLOW



Also maintained an admin panel built with **react-admin**.

2018.07 - 2021.04



Nwave Technologies Ltd.

At Nwave, an IoT company building smart parking sensors for UK clients, I worked as a Full Stack Web Developer on both frontend and backend systems managing **20,000 IoT devices**.

Built a **REST API** service handling full CRUD operations for 20,000 devices, achieving **99% test coverage**, by implementing it with **AWS Lambda, Node.js/TypeScript, PostgreSQL (PostGIS), AWS API Gateway**, and **AWS Cognito**, tested with **Mocha, Chai**, and the **AWS SDK**. **↑ RELIABILITY**

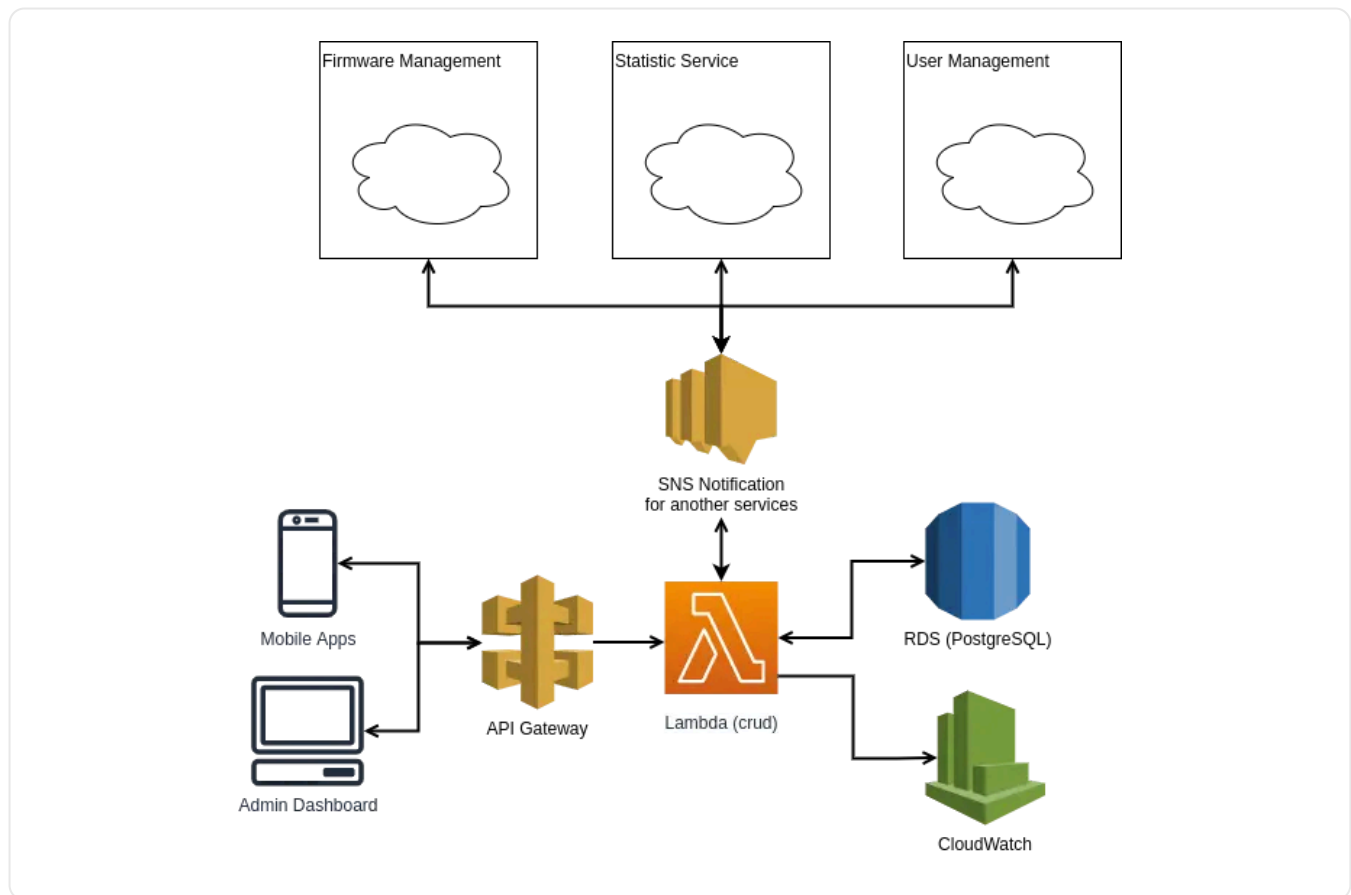
Improved infrastructure stability and enabled repeatable environment deployments, measured by one-command infrastructure setup, by adopting **AWS CDK** as infrastructure-as-code — replacing manual cloud configuration with version-controlled, reproducible stacks. **↑ DEVOPS**

Improved rendering performance of 20,000 map markers in the **SPlace PWA** mobile app, eliminating lag on low-end devices, by implementing marker clustering, removing unnecessary recalculations, and optimizing the **Ionic/Angular** rendering pipeline.

↑ PERFORMANCE

Also developed and maintained an admin dashboard for monitoring sensor statuses, built with **React, Redux-Saga, TypeScript, Material-UI, AWS Amplify**, and the **Google Maps API**. The app had full unit and end-to-end test coverage, integrated with **Bitbucket CI/CD** and **Sentry**.

ARCHITECTURE DIAGRAM



2017.05 - 2018.06



At AdGuard, one of the most popular ad blockers, I contributed to both the main product and additional browser extensions, written in **JavaScript (ES6)**:

- An extension that helps users read website content hidden behind ad-blocker detection walls (not deployed, available in a [repository](#)).
- Contributions to **AdGuard Assistant** ([repository](#)).

Also worked on the front-end of the [AdGuard](#) website using **Vue**, **Vuex**, and **PostCSS**.

2015.05 - 2017.05

Worked as a Web Developer for several companies, building and maintaining websites using **PHP**, **WordPress**, **JavaScript**, and **jQuery**. Started working with modern JavaScript frameworks, including **AngularJS** (Angular 1), and built single-page applications. Gained

experience with real-time features using **WebSockets** and created interactive **SVG** animations for various projects. Automated build and deployment processes using **Grunt** and **Gulp**, which improved development consistency across environments.